YUNA SIM

Berlin, Germany | syna02178@gmail.com | +49 1520 5820897 | LinkedIn

PROFESSIONAL SUMMARY

AI/ML researcher with an M.S. in Artificial Intelligence and research experience in vision-language-action (VLA) models, computer vision and multispectral imaging. At Nota AI, reproduced a 3D-aware VLA policy and found defects that had invalidated experiment results. 4 publications (3 first-author) and 2 Korean patent applications.

PROFESSIONAL EXPERIENCE

NOTA AI Berlin, Germany | Jul 2026 – Oct 2026

AI Research Intern

- Reproduced a VGGT-based 3D-aware VLA policy (VGA) as the team baseline: reconciled the paper with the official VGGT code, resolved unspecified details with the authors, and validated components against the OpenVLA-OFT implementation
- Identified and fixed result-invalidating defects, including a language-embedding cache mismatch caused by a library version difference, a depth-token ordering bug (now covered by regression tests) and evaluation-order dependence in LIBERO
- Measured evaluation variance across training seeds, sampling seeds and checkpoints before comparing methods, and fixed decision criteria in git before viewing results
- Added per-step rollout recording to LIBERO evaluation and used replay and interventions on approach pose and object layout to localize manipulation failures to the pre-contact approach
- Ported a discrete diffusion action decoder and ran future-depth auxiliary and geometry-conditioning experiments, using input-ablation diagnostics to test whether added 3D signals changed predicted actions

ENERDOT Seoul, Korea | May 2026 – Jul 2026

Data Science Intern

- Reduced day-ahead Jeju SMP forecast MAE by ~14% versus a 7-day persistence baseline (14.52 → 12.53 KRW/kWh) on a two-month held-out test set
- Built Airflow/MySQL ETL and MLflow-tracked forecasting pipelines, and corrected optimistic early solar results caused by shuffled splits and a train period overlapping the evaluation window

Computational Media Research Seoul, Korea | May 2024 – Feb 2025

Research Assistant, Ewha Womans University | *Advisor: Prof. Hoyoung Yoon*

- Analyzed HSV/LAB color characteristics of Korean dramas to study viewer color preferences and emotional engagement
- Designed viewer surveys and ran the statistical analysis for a first-author paper in the Journal of the Korean Data Analysis Society

Information Coding and Processing Laboratory (ICPL) Seoul, Korea | Sep 2022 – Dec 2024

Research Assistant, Ewha Womans University | *Advisor: Prof. Jewon Kang*

- Proposed SPSNet, a spectral patch-selection framework for global illuminant estimation from 15-band multispectral images, trained on angular-error pseudo-labels from a pretrained estimator (M.S. thesis)
- Reduced mean angular error by 6–9% versus the CCNet baseline across four datasets (1,401 images), with consistent gains in worst-25% error
- Co-authored two KIBME papers and two Korean patent applications on illumination estimation and multi-view auto white balance

Innovation Technology Laboratory (ITL) Seoul, Korea | Jan 2023 – Jun 2023

AI Research Intern

- Trained a YOLOv5s weld-defect detector reaching 87.5% mAP@0.5 through two-stage transfer learning, +9.1 pp over training without transfer
- Built a CycleGAN pipeline converting grayscale radiographic weld images into RGB camera-style training data; first-authored the IPIU 2024 paper

PROJECTS

Welfare Compass Seoul, Korea | Nov 2025 – Dec 2025

Seoul AI Hackathon | *Finalist, top 20 of 181 teams*

- Led ideation and built the core RAG retrieval/generation logic for a welfare-policy assistant covering 300+ programs (LangChain, ChromaDB, GPT API, Streamlit)

PUBLICATIONS AND PRESENTATIONS

- Yuna Sim, Hoyoung Yoon, **"A Preference Analysis of Color: A Pilot Study Comparing the Drama <Money Heist> Original and Remake Versions"**, Journal of the Korean Data Analysis Society, Feb. 2025
- Yuna Sim, Hyejin Oh, Jewon Kang, **"Global Illumination Estimation Using Local Patch Selection"**, The Korean Institute of Broadcast and Media Engineers (KIBME), Nov. 2024
- Hyejin Oh, Yuna Sim, Jewon Kang, **"Deep Learning-Based Fast Auto White Balancing Algorithm Development for Multi-View Video"**, The Korean Institute of Broadcast and Media Engineers (KIBME), Jun. 2024
- Yuna Sim, Jewon Kang, **"Welding Defect Detection Using Synthetic Data and Object Recognition Algorithms"**, Image Processing and Image Understanding (IPIU), Feb. 2024

PATENT APPLICATIONS

- **Global Illumination Estimation Method Based on Local Illumination Estimation and Image Processing Apparatus**, Yuna Sim, Hyejin Oh, Jewon Kang, Korean Patent Application No. 10-2024-0164675 (filed 2024, pending)
- **Auto White Balancing Method for Multi-View Video and Image Processing Apparatus**, Yuna Sim, Hyejin Oh, Jewon Kang, Korean Patent Application (filed 2024, pending)

EDUCATION

Ewha Womans University Seoul, Korea | Sep 2022 – Feb 2025

M.S. in Artificial Intelligence Convergence

Thesis: Global Illuminant Estimation Using Local Patch Selection with Multispectral Imaging

Ewha Womans University Seoul, Korea | Mar 2016 – Aug 2022

B.S. in Electronics Engineering, B.A. in Multimedia (Double Major)

SKILLS

ML / CV: PyTorch, Transformers/ViT, CNN, YOLOv5, CycleGAN, VLA (VGGT-based), LIBERO, ablation design

Programming & data: Python, SQL, pandas, scikit-learn, LightGBM, unit testing, Git, Docker, Airflow, MLflow

LLM: LangChain, LangGraph, ChromaDB, RAG

Languages: Korean (native), English (advanced, OPIc AL), German (A1)